

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech.

B ranch : E&TC Engineering

Semester : V

Subject Code & Name: BTEXC505 Microcontroller & its Applications

Max Marks: 60

Date: 19/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve any two of the following.		12
A) Distinguish between RISC & CISC processors.	L6	6
B) Briefly explain TMOD register of 8051 microcontroller.	L3	6
C) Provide comprehensive explanation of internal RAM structure of 8051.	L3	6
Q.2 Solve any two of the following.		12
A) Draw pin diagram of 8051 microcontroller and explain functions of any four pins of it.	L6	6
B) State which timer and which mode is selected for following values of TMOD register. i. TMOD = 10H ii. TMOD = 11H iii. TMOD = 21H	L4	6
C) State which interrupts are enabled for following values of IE register. i. IE = 88H ii. IE = 99H iii. IE = 7FH	L5	6
Q. 3 Solve any two of the following.		12
A) Write an 8051 assembly or C program to transmit message “ABC” serially with 9600 baudrate.	L6	6
B) Briefly explain functions of any six pins of an RS-232 DB 9 connector.	L6	6
C) Provide comprehensive explanation of functions of all pins of 16x2 LCD device.	L5	6

Q.4 Solve any two of the following.		12
A) Write an 8051 C program to rotate stepper motor in clockwise direction continuously.	L4	6
B) Provide comprehensive explanation of Status Register of a PIC 18 microcontroller.	L1	6
C) Explain any three addressing modes of PIC 18 microcontroller with suitable examples of each.	L3	6
 Q. 5 Solve any two of the following.		12
A) Briefly explain functions of registers associated with PIC 18 I/O ports.	L1	6
B) Explain following instructions of PIC 18 with suitable examples of each.	L5	6
i. MOVLW		
ii. BCF		
iii. ADDLW		
C) Write a Pic 18 assembly language program to toggle all pins of PORT B.	L3	6

***** End *****